Differentiable transmon design with a tensor-native finite-element electromagnetics solver: gradient-based optimization of charging energy, cross-validated against Palace

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Abstract

Superconducting-qubit electromagnetic design is an iterative guess-and-check loop: the workflow's production field solvers expose no derivatives of design figures-of-merit, so state-of-practice optimizers iterate time-consuming electromagnetic simulations and inject parameter updates from separate analytic physics models (Eriksson et al., 2025). We demonstrate the missing capability — *solver-derived* design gradients from a 3D finite-element electromagnetics solve — and use it for gradient-based optimization of a transmon's charging energy E_C. GEODE-FEM is a Rust solver built on the Burn tensor framework; a discrete-adjoint layer bridges the one stage its autodiff tape cannot reach (the sparse factorization), yielding a validated 2×2 sensitivity matrix — material ($\partial/\partial\varepsilon$) and geometry ($\partial/\partial X$) derivatives on both the electrostatic and the driven H(curl) operators — each finite-difference-validated with mutation tests (relative errors 10^{-9}–10^{-5}). Because the capacitance $C = \phi^{\mathsf{T}} K \phi$ is variationally stationary, its shape derivative collapses to a Hellmann–Feynman-like explicit term; composing it with the capacitance-to-E_C chain, Newton iteration drives a capacitor geometry to a 0.2156 GHz target E_C, confirmed by an independent fresh solve to 1.4×10^{-15}, and on the real 133k-tetrahedron device mesh $\partial C_\Sigma/\partial\theta$ is FD-validated at 1.15×10^{-4} with a genuine two-step convergence to a within-budget target. The honest negatives are stated, not buried: the fixed-topology pad parametrization falls 33× short of the ∼90 fF design anchor (a mesh-morphing problem, named as future work), and eigenmode/EPR derivatives remain roadmap, blocked on a characterized deep-shift eigensolve wall. The correctness credential: on the identical mesh, GEODE-FEM reproduces all six of Palace's transmon eigenmodes to $\leq 0.033\%$ (worst case 0.032%). The contribution is a capability, not a speed race — on raw performance the measured record shows no corner where GEODE-FEM beats Palace at scale, and we report it honestly. Every number traces to a committed artifact.

1 Introduction

The hard problem in superconducting-qubit electromagnetic design is the *inverse map*: given target Hamiltonian parameters — a charging energy E_C, an anharmonicity α, coupling capacitances — produce a physical layout. Today that map is walked by hand: parameterize a geometry, run a 3D electromagnetic simulation, extract the parameters, compare, adjust, repeat. The state-of-practice automated workflow makes the loop's structure explicit: QDesignOptimizer (Eriksson et al., 2025) iterates time-consuming electromagnetic simulations of HFSS-class solvers and guides the parameter updates with *separate*, user-defined analytic physics models — precisely because the field solver at

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the center of the loop produces observables but no derivatives of those observables with respect to the design. The gradient the optimizer actually wants, ∂(figure of merit)/∂(geometry, materials) through the field solve itself, is structurally unavailable from the production solvers — commercial (HFSS, COMSOL) and open-source (Palace (Palace developers, AWS Center for Quantum Computing, 2023)) alike: their assembly and factorization pipelines are hand-written kernels with no derivative path, and none exposes an adjoint of its solve.

The surrounding literature brackets this gap without closing it. At the *lumped-circuit* level, differentiation is a solved problem: SQcircuit (Rajabzadeh et al., 2023) analyzes arbitrary superconducting circuits, and its gradient-based follow-up (Rajabzadeh et al., 2024) computes eigenvalue and eigenvector gradients of the circuit Hamiltonian via PyTorch autodiff, optimizing qubit designs in circuit-parameter space. But the map from geometry and materials *to* those lumped parameters still runs through non-differentiable 3D electromagnetic simulation. At the *field* level, differentiable electromagnetics is owned by photonics: adjoint and autodiff methods on FDTD and integral-equation solvers are a mature inverse-design industry (Hammond et al., 2022; Hughes et al., 2019; Mahlau et al., 2026; Molesky et al., 2018; Ponomareva et al., 2025; Schubert et al., 2022), and differentiable FEM at scale has been proven outside electromagnetics by JAX-FEM in solid mechanics (Xue et al., 2023). What none of these supplies is the regime superconducting-qubit (and, more broadly, RF) design lives in: *frequency-domain finite elements on body-fitted tetrahedra with anisotropic materials* — distributed, field-level EM gradients from the same FEM solve that produces the design observables. That wedge is, to our knowledge, uncontested; this paper occupies it.

GEODE-FEM ("geode-fem") is positioned to close the gap by construction rather than retrofit. It is a full-wave 3D $H(\text{curl})$/Nédélec (plus scalar electrostatic) finite-element solver written in Rust on the Burn tensor framework (Tracel AI, 2024): element assembly is batched $[n_{\text{elem}}, 6, 6]$ tensor algebra on an autodiff-capable substrate, so the derivative of assembly with respect to materials and node coordinates is available where a classical solver has only hand-written kernels. One stage breaks the tape: the sparse direct factorization (faer (Kazdadi, 2026)) at the heart of every solve. A discrete-adjoint layer closes exactly that gap — one extra (transpose-)solve, reusing the forward factorization, restores the full gradient (Section 8). The resulting capability is general — shape and material sensitivities for RF, photonic, and electrostatic device design — and the transmon is our demonstration vehicle because its unmet need is the clearest documented in the literature, not because the method is qubit-specific.

A design-gradient claim from a new solver is only as credible as the solver's forward answers, so the paper carries its correctness credential up front: on a real transmon-plus-readout-resonator geometry meshed once and consumed identically by both codes, geode-fem reproduces Palace — the MFEM-based (Anderson et al., 2021; Andrej et al., 2024) reference solver for superconducting-qubit design — to within 0.033% across all six eigenmodes (worst case 0.032%; junction LC mode 0.001%), with exact analytic tripwires guarding against coincidental agreement (Section 5). This parity is *a credential, not the contribution* — matching Palace is the price of entry — and it is, incidentally, the first independent third-party validation of the Palace/DeviceLayout.jl transmon stack, which has no publication of its own.

Our contributions are:

- **A validated 2×2 discrete-adjoint sensitivity matrix for FEM electromagnetics.** Material ($\partial/\partial\varepsilon$) and geometry ($\partial/\partial X$, exact $\partial K/\partial X$ via forward-mode duals through the element kernels) derivatives, on both the scalar electrostatic and the complex driven $H(\text{curl})$ Nédélec operators — each one forward plus one adjoint solve sharing a single factorization, each validated against full-pipeline central finite differences with mutation tests that reject sign, conjugation, and dropped-term errors (Section 8).

- **A differentiable capacitance→ E_C chain with a variational-stationarity simplification.** Because $C = \phi^{\mathsf{T}} K \phi$ is stationary in the solved potential, the implicit term vanishes and the shape derivative is a pure explicit-geometry (Hellmann–Feynman-like) term; validated against the analytic parallel-plate law and central finite differences (Section 8.2).

- **Gradient-based optimization demonstrations, honest tiers labeled.** An end-to-end Newton loop that hits a 0.2156 GHz target E_C and is confirmed by an independent fresh solve at 1.4×10^{-15} (the parametrization is affine, so one-step convergence proves the loop, not a hard optimization); a real-device demonstration on the committed 133k-tet mesh with the gradient FD-validated at 1.15×10^{-4} and a genuine two-step convergence; and the honest negative — the fixed-topology pad parametrization falls 33× short of the ∼90 fF design anchor, a mesh-morphing problem we name rather than hide (Section 9).

- **The correctness credential.** Same-mesh, same-physics, same-junction-model cross-validation against Palace at $\leq 0.033\%$ across all six modes, with oracle-first methodology and honest-physics reporting — the by-construction absent qubit mode and a disclosed, diagnosed spurious mode (Sections 5 and 6) — doubling as the first independent validation of the Palace/DeviceLayout.jl stack.

- **Honest performance and roadmap accounting.** A matched CPU cell shows a per-core, small-to-medium-scale win for the direct eigensolve that *loses at scale on both axes* — at ∼1.16M DOFs it completes given adequate memory (565.5 s, 92.2 GB peak) but Palace is faster and far leaner (423.12 s, ∼33 GB aggregate); a flop-and-fill crossover below 1M DOFs, not merely a memory wall — and the GPU cell is correctness-proven, performance-negative at measured sizes (Section 11). Eigenmode/EPR derivatives are *not yet differentiable* here; the blocker and the named path are stated as roadmap, with no overclaim (Section 10).

Section 2 positions the work; Section 3 presents the substrate and where its tape breaks; Sections 4–6 establish the benchmark problem and the correctness credential; Sections 7–9 are the contribution — the forward LOM chain, the adjoint layer, and the optimization demonstrations; Section 10 is the eigenmode roadmap; Sections 11–13 report performance context, reproducibility, and discussion.

2 Related Work

Superconducting-qubit design optimization. The design loop this paper attacks is documented across the qubit-design software stack. QDesignOptimizer (Eriksson et al., 2025) is the state of practice: physics-guided multi-parameter optimization that wraps non-differentiable EM simulation (HFSS with pyEPR/Qiskit-Metal-class post-processing) and steers it with analytic models, because no gradient is available from the solver. The SQcircuit line (Rajabzadeh et al., 2023, 2024) differentiates at the lumped-circuit level — including eigenvalue/eigenvector gradients of the circuit Hamiltonian via PyTorch — and optimizes in circuit-parameter space; the geometry-to-parameters map remains simulation-bound. The SQuADDS database (Shanto et al., 2024) attacks the same loop by lookup: pre-simulated, experimentally-validated design points. Qiskit Metal's quasi-lumped methods (Minev et al., 2021b) and the EPR/black-box quantization physics (Koch et al., 2007; Minev et al., 2021a; Nigg et al., 2012) frame the parameter-extraction layer this paper's LOM chain implements. None of these obtains gradients from the field solve itself; that is the gap Section 8 closes.

Differentiable electromagnetics — owned by photonics. Adjoint/autodiff EM is a mature inverse-design discipline in nanophotonics (Molesky et al., 2018), expressed through structured grids and integral methods: autograd FDFD/FDTD in ceviche (Hughes et al., 2019), foundry-constrained JAX inverse design (Schubert et al., 2022), the Meep adjoint pipeline (Hammond et al., 2022), the JAX-native FDTDX framework (Mahlau et al., 2026), GPU-accelerated autodiff integral-method scattering in TorchGDM (Ponomareva et al., 2025), and differentiable RCWA (Kim et al., 2024). These codes accept rectilinear grids or Green's-function discretizations; the superconducting-qubit regime — body-fitted tetrahedra, anisotropic sapphire, $H(\mathrm{curl})$ conformity, frequency domain — is exactly where that literature stops. Differentiable FEM has been proven outside EM: JAX-FEM (Xue et al., 2023) (GPU-accelerated, autodiff-native continuum mechanics) and PyTorch-FEA (Liang et al., 2023) in biomechanics; concurrent work on general differentiable Galerkin assembly (TensorGalerkin (Wen et al., 2026), with the torch-sla adjoint sparse-solver library (Chi and Wen, 2026)) independently confirms the substrate-level thesis while reporting nodal magnetostatics rather than full-wave $H(\mathrm{curl})$ EM. To our knowledge, no prior work delivers field-level design gradients from a frequency-domain 3D FEM electromagnetics solve of the class RF/superconducting design uses.

FEM kernels by code generation, and tensor compilers. Generating FEM kernels from high-level descriptions is a two-decade tradition — FFC (Kirby and Logg, 2006), UFL (Alnæs et al., 2014), Firedrake (Rathgeber et al., 2017), TSFC (Homolya et al., 2018) — whose modern GPU expression is libCEED (Brown et al., 2021) and MFEM's partial-assembly path (Andrej et al., 2024). Palace itself runs on that domain-specific stack. geode-fem's substrate bet is categorically different: outsource kernels, scheduling, and backend portability to a *general-purpose ML tensor stack* (the Halide/TVM compiler lineage (Chen et al., 2018; Ragan-Kelley et al., 2013), reaching scientific computing via ML frameworks), and phrase assembly and matrix-free application as batched tensor algebra. For this paper the decisive property of that choice is not portability but *differentiability*: an autodiff-capable substrate makes the assembly derivative available by construction, which is what the adjoint layer of Section 8 composes with. Adjacent genera — simulation DSLs (Holl et al., 2020; Hu et al., 2019), neural-ansatz eigensolvers (Jiang et al., 2024), hand-written CUDA FEM (Meng et al., 2014) — trade away either the general-purpose stack or the discretization itself.

Validation and cross-solver benchmarking. Code-to-code comparison on precisely specified problems is an established genre in computational electromagnetics (the TEAM workshop tradition since 1986[1]; nano-optics solver comparisons (Hoffmann et al., 2009)); in V&V vocabulary (Oberkampf and Roy, 2010; Roache, 1997) our credential is verification-side, code-to-code evidence. On the qubit-EM axis the closest works are SQDMetal (Sommers et al., 2025) (Palace cross-checked against COMSOL/HFSS at the 0.02–0.13% level, independently constructed models) and the Palace-centered workflow of Ye et al. (2025) (validated against cryogenic measurement at $\sim$0.3%). Both wrap and trust Palace; neither is a same-mesh comparison against an independently *implemented* solver. Our protocol eliminates meshing and model entry as confounders — identical MSH fixture, identical lumped-shunt junction, first-order Nédélec on both sides — so residuals isolate formulation and solver correctness. Mature open EM tools normally carry a methods paper (GetDP (Dular et al., 1998), FEniCS/DOLFIN (Logg and Wells, 2010), Gmsh (Geuzaine and Remacle, 2009), scikit-rf (Arsenovic et al., 2022), MFEM (Anderson et al., 2021)); Palace and DeviceLayout.jl are unpublished outliers (AWS Center for Quantum Computing, 2025; AWS Quantum Technologies Blog, 2023; Palace developers, AWS Center for Quantum Computing, 2023; Peairs, 2025; Peairs and Carson,

[1] The TEAM founding documents are workshop reports without resolvable DOIs; we describe the tradition in prose.

2025), which is the gap the credential narrows from the outside. geode-fem's numerics: first-order Nédélec elements (Nédélec, 1980), shift-invert Lanczos (Lehoucq et al., 1998) over a real symmetric pencil factorized with faer (Kazdadi, 2026); Palace: SLEPc-class Krylov (Hernandez et al., 2005) with divergence-free projection and AMS preconditioning (Hiptmair and Xu, 2007; Kolev and Vassilevski, 2009); the gauge-fixing literature behind Section 6 is the tree–cotree tradition (Albanese and Rubinacci, 1988; Manges and Cendes, 1995).

3 GEODE-FEM: A Tensor-Native Substrate, and Where the Tape Breaks

geode-fem is written in Rust on the Burn tensor framework (Tracel AI, 2024), whose cubecl JIT compiles tensor kernels for CPU, CUDA, and WebGPU backends from one source. Every throughput-critical FEM stage is phrased as the batched dense algebra the stack already optimizes.

Batched element-local assembly. For first-order Nédélec tetrahedra the element curl–curl and mass contributions are 6×6 matrices depending on per-element geometry factors and material tensors (including the rotated anisotropic sapphire permittivity of Section 4.1); the scalar electrostatic operator is the analogous P1 kernel. geode-fem computes them for all elements at once as batched $[n_{\text{elem}}, 6, 6]$ tensor operations — one kernel launch per assembly stage rather than an element loop — then either scatters into a global sparse matrix (the assembled f64 CPU path) or retains the batch for matrix-free application.

Matrix-free application and on-device Krylov. The matrix-free operator apply is a three-stage tensor pipeline — gather global DOFs to element-local vectors, batched matmul against the stored element operators, scatter-add back — and the driven frequency-sweep solve runs a conjugate orthogonal conjugate gradient (COCG) iteration entirely on the device with a constant number of scalar host–device synchronizations per iteration. The backend (ndarray on CPU, CUDA on GPU) is a type parameter, not a code path. Section 11 reports honestly how far below the GPU-amortization threshold the current benchmark sizes sit.

Honest constraints. Two boundaries are constraints, not footnotes. The CUDA backend (burn-cuda 0.21) is f32-only today — the underlying cubecl runtime currently disables f64[2] — so every GPU result is mixed-precision-qualified. And the eigensolve that produces the credential comparison is factorization-bound: its inner solves are faer sparse-LU factorizations (Kazdadi, 2026), a classical serial CPU component outside the tensor-compiler story.

Where the tape breaks — and why an adjoint layer, not autodiff-through-the-solver. The substrate choice has a consequence the rest of this paper builds on: Burn's automatic differentiation reaches everything *up to* the linear solve — material and node-coordinate dependence of the assembled operators is differentiable by construction — but the sparse factorization is an external, opaque kernel that breaks the tape. Differentiating *through* a factorization by unrolling is neither practical nor necessary: the classical discrete-adjoint identity turns the missing derivative into one additional linear solve against the *same* factorization (Section 8). The division of labor is deliberate — autodiff where the substrate provides it (element kernels, via forward-mode duals), adjoint algebra where it does not (the solve) — and it is what makes the gradients in this paper *solver-derived* in the sense the design loop of Section 1 lacks: the same assembly and the same factorization that produce the observable produce its gradient.

[2] Upstream constraint of the burn/cubecl stack at version 0.21 (Tracel AI, 2024); tracking-issue URL to be added at revision. TODO(operator).

4 Benchmark Problem: Geometry, Mesh, and Junction Model

4.1 Geometry and mesh provenance

The device is the `SingleTransmon` example shipped with DeviceLayout.jl v1.15.0 (AWS Center for Quantum Computing, 2025): a transmon with readout resonator on a sapphire substrate, enclosed in a vacuum box. The sapphire is anisotropic, with permittivity tensor $\varepsilon = R\,\mathrm{diag}(9.3,\, 9.3,\, 11.5)\, R^{\mathsf{T}}$ rotated approximately 36.87° in-plane. The DeviceLayout.jl workflow emits seven named physical groups (metal, substrate, vacuum, junction lumped-element surface, two readout ports, exterior boundary) and drives Gmsh (Geuzaine and Remacle, 2009) to produce a tetrahedral mesh, exported as an MSH 4.1 fixture and pinned by SHA-256 (Section 12).

The mesh has 22,684 nodes and 133,314 tetrahedra, yielding 156,863 Nédélec edge unknowns, of which 133,108 interior degrees of freedom remain after eliminating perfect-electric-conductor (PEC) boundary edges. PEC is applied on all metal surfaces and the exterior box; the two readout ports are left open (lossless), so no resistive element enters and the eigenproblem pencil stays real symmetric. Figure 1 shows the geometry with physical groups color-coded. Every measurement in this paper — the eigenmode credential, the electrostatic LOM chain, and the real-device gradient demonstration — runs on this one committed mesh (plus a uniformly refined variant for the scale study of Section 11).

4.2 The junction model: a lumped reactive shunt

The Josephson junction is modeled identically in both solvers as a lumped reactive shunt — a parallel $L \parallel C$ element with $L = 14.860\,\mathrm{nH}$ and $C = 5.5\,\mathrm{fF}$ — attached to the four-triangle `lumped_element` surface group Γ. Following the uniform lumped-port construction, with port length ℓ and width w, the shunt contributes

$$K_{\mathrm{port}} = \frac{\ell}{w\,L}\, S_\Gamma \qquad \text{and} \qquad M_{\mathrm{port}} = \frac{C\,\ell}{w}\, S_\Gamma, \tag{1}$$

where S_Γ is the port-surface matrix assembled on Γ. The eigenproblem is then the real symmetric generalized pencil

$$(K + K_{\mathrm{port}})\, x \;=\; \omega^2\, (M + M_{\mathrm{port}})\, x. \tag{2}$$

In Palace, the same physics is entered as a reactive `LumpedPort` ($L \parallel C$, no resistance) on the same surface group; the readout ports are configured open, so Palace's pencil is likewise real. This one-to-one correspondence is what makes the comparison same-physics as well as same-mesh.

5 The Correctness Credential: Same-Mesh Cross-Validation Against Palace

A solver claiming design gradients must first prove its forward answers; this section is that proof. Palace's own agreement with commercial tools is established (Sommers et al., 2025), so reproducing Palace is the bar a new implementation must clear before anything it computes *about* its solutions (such as their derivatives) is worth discussing.

Protocol. The benchmark is oracle-first. The mesh is generated once, committed, pinned by SHA-256, and consumed by both solvers at matched (first) order with the identical junction model; residual disagreement is therefore attributable to formulation or implementation, not meshing. Two exact analytic tripwires guard against coincidental or self-consistent-but-wrong agreement: the isolated junction LC resonance $f_{LC} = 1/(2\pi\sqrt{LC}) = 17.60\,\mathrm{GHz}$ anchors the junction mode, and a Josephson $\sqrt{L}$-scaling test requires that doubling L move the junction mode by exactly $1/\sqrt{2}$ — measured: $17.4901 \to 12.37\,\mathrm{GHz}$, committed ratio 0.7071 (computed from the unrounded values; Figure 3). The suite also contains inverse tests that *must fail* (perturbed configurations required

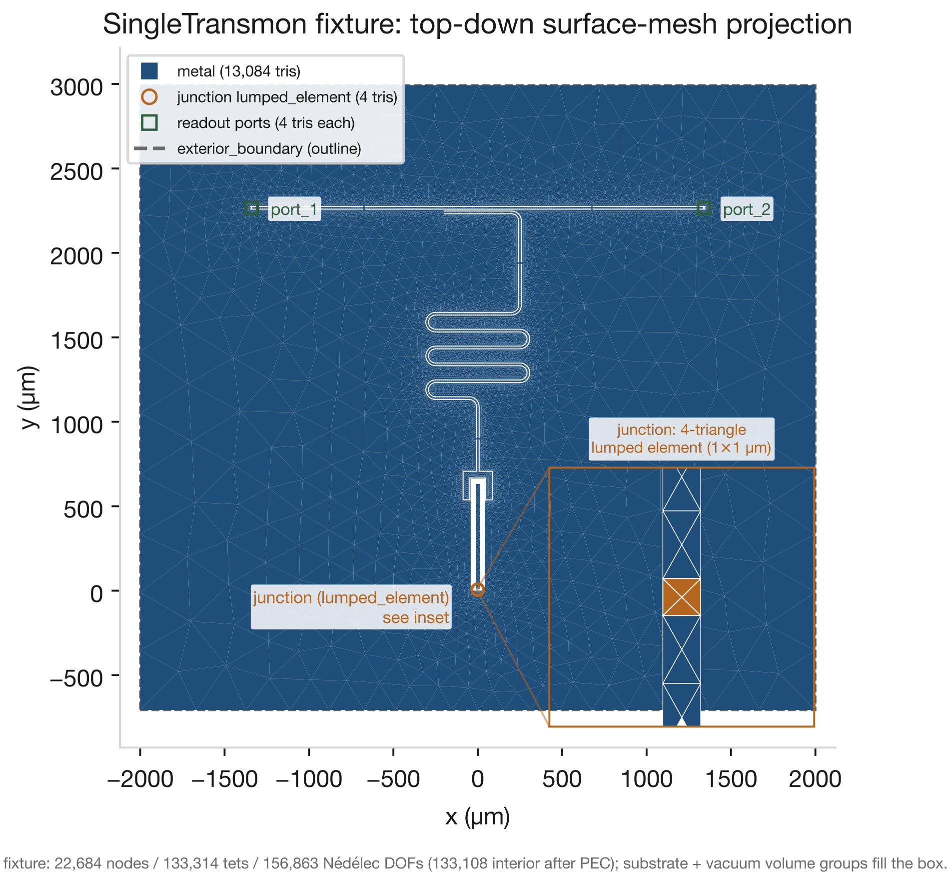

Figure 1: Benchmark geometry: top-down projection of the committed surface mesh (the SHA-256-pinned MSH 4.1 fixture) for the DeviceLayout.jl v1.15.0 `SingleTransmon` transmon and readout resonator. Surface physical groups are color-coded — the metal ground plane (with the coplanar-waveguide feedline, meander resonator, and transmon slot as etched gaps), the four-triangle junction `lumped_element` surface (enlarged in the inset, actual triangulation shown), the two readout ports, and the exterior-boundary outline; the `substrate` and `vacuum` volume groups fill the enclosing box. Both solvers consume this identical mesh (22,684 nodes; 133,314 tetrahedra).

to disagree with the oracle — a pipeline that still passes under them is not measuring anything), and the Palace oracle is deterministic: a rerun from the committed configuration reproduced its committed `eig.csv` bit-for-bit.

Mode identification. Modes are identified, not eyeballed. Within geode-fem, the stiffness-participation ratio

$$p \;=\; \frac{x^{\mathsf{T}} K_{\text{port}}\, x}{x^{\mathsf{T}}\, (K + K_{\text{port}})\, x} \tag{3}$$

cleanly separates the junction LC mode ($p = 1.000$) from resonator and box modes ($p = 0.000$ at the artifact's three-decimal precision). Palace reports a field-based per-port energy participation, a differently normalized quantity that does *not* rank the modes the same way on the committed run (all six of its port-EPR values are $\leq 5 \times 10^{-4}$, with the junction mode's the smallest in magnitude); the two diagnostics are complementary, not a shared discriminant, and we do not compare them numerically. Cross-solver mode identification therefore rests on frequency agreement against the

Table 1: Eigenmode agreement on the identical mesh (22,684 nodes / 133,314 tets / 156,863 Nédélec DOFs, 133,108 interior after PEC), first order in both solvers, junction as a lumped reactive shunt ($L = 14.860$ nH, $C = 5.5$ fF). geode-fem at commit `3174015`; Palace at commit `fba6a5b`. All values are quoted verbatim from the committed `benchmarks/transmon_eigen/results.toml`: geode-fem frequencies at the committed three decimals, Palace at full precision, and the committed per-mode `rel_err_pct`, computed against the rounded geode-fem values (precision disclosure in Section 12).

Mode		geode-fem (GHz)	Palace (GHz)	Δ (%)
`resonator`	readout resonator	5.153	5.151335830348	0.032
`mode_2`	box/cavity	15.465	15.46052107794	0.029
`junction_lc`	junction LC	17.490	17.49010903536	0.001
`mode_4`	box/cavity	18.693	18.69165792915	0.007
`mode_5`	box/cavity	20.703	20.69755679425	0.026
`mode_6`	box/cavity	26.088	26.08089940472	0.027

analytic anchor.

Result. Table 1: all six modes agree to $\leq 0.033\%$ (worst case 0.032%), with the junction LC mode — the mode the junction model exists to produce — at 0.001%. The junction mode sits 0.6% below the isolated-circuit anchor $f_{LC} = 17.60$ GHz in *both* solvers (the small shift reflects its embedding in the field problem), and the agreement is between genuinely independent code paths — a batched-tensor assembly feeding a direct-factorization Lanczos in Rust, and a projected, preconditioned Krylov method in C++ — so a common implementation error is an unlikely explanation for the residual, and the tripwires exclude self-consistent-but-wrong junction physics.

6 Honest Physics: The Absent Qubit Mode and a Disclosed Spurious Mode

6.1 No $\sim$4 GHz qubit mode exists, by construction

The DeviceLayout.jl blog workflow (Peairs and Carson, 2025) reports a spectrum running from 4.14 to 5.591 GHz — including a $\sim$4 GHz qubit mode — while Table 1 contains no mode below 5.15 GHz. This is a property of the model, stated plainly: the physical transmon frequency arises from the Josephson inductance resonating against the *pad (shunt) capacitance* of roughly 80–100 fF (Koch et al., 2007), and the lumped-shunt model of Section 4.2 gives the junction only its own 5.5 fF, so its resonance lands at 17.5 GHz. Both solvers agree on the absence: Palace's projected eigensolve, instructed to hunt at $\sigma = 4.5$ GHz, finds nothing below the 5.15 GHz resonator. The blog spectrum is not reproduced by this model, and no agreement with it is claimed. The route to qubit quantities from this geometry is the electrostatic LOM chain of Section 7 — which is exactly the chain this paper makes differentiable.

6.2 One spurious mode: disclosed, diagnosed, characterized

Honest cross-validation requires reporting where the solvers *disagree*, and there is exactly one such place. geode-fem's headline Lanczos operates on the raw pencil of Eq. 2 without a gauge projection, and on this problem it leaks one spurious mode at 3.4528 GHz, junction-localized ($p = 0.9942$), with no Palace counterpart. The artifact was run to ground through a measured three-step gauge/projection arc, every step committed (Section 12): (1) classical tree–cotree elimination (Albanese and Rubinacci, 1988; Manges and Cendes, 1995) removes it but is not spectrum-preserving for the generalized eigenproblem — the resonator shifts 1.64% (5.1528 $\rightarrow$ 5.2372 GHz), outside our bar, pinned as a regression test; (2) the spectrum-preserving bulk divergence-free projector

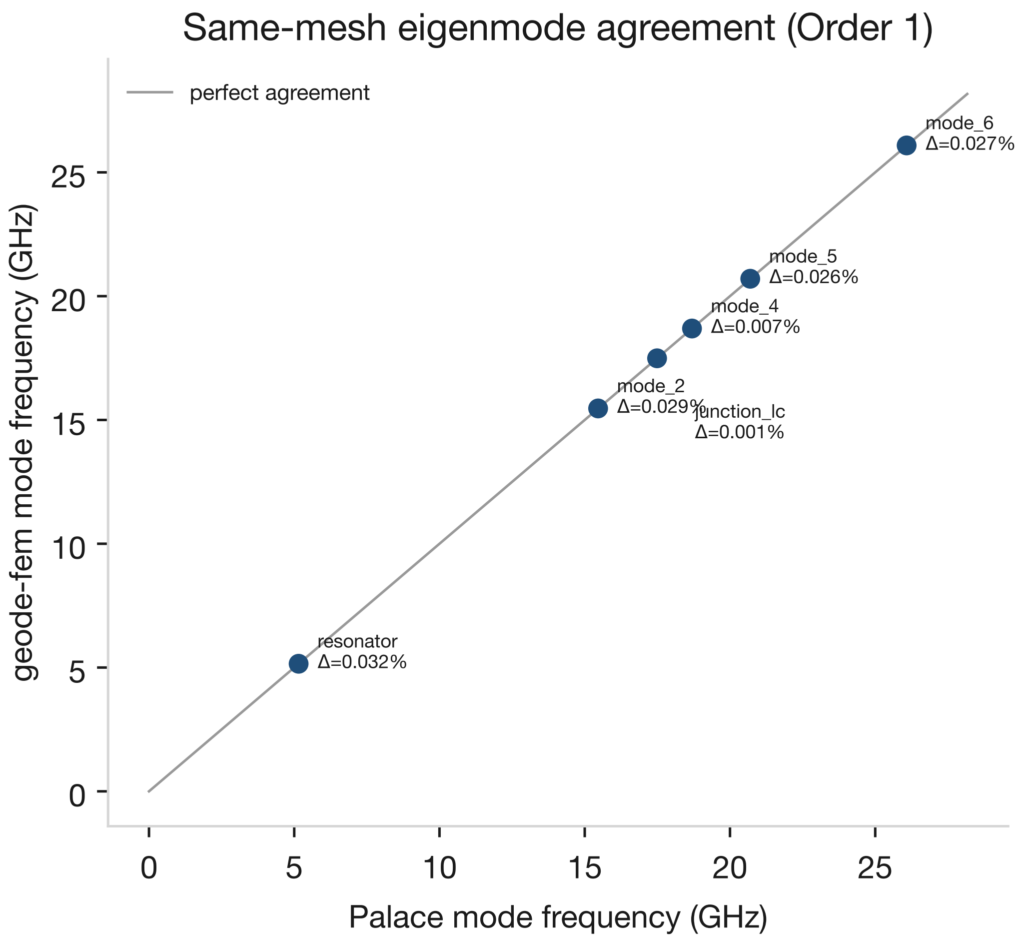

Figure 2: Mode-frequency agreement: geode-fem versus Palace on the identical mesh, with per-mode relative deviations annotated. Data from the committed benchmark artifacts (`benchmarks/transmon_eigen/results.toml` and Palace's committed `eig.csv`).

(structurally Palace's confinement) preserves the cavity spectrum (0.029%) and collapses the 13,747-mode gradient cluster, but *deflates the physical junction mode* — a lumped inductor is a quasi-static, curl-free flux path (measured divergence ratio 50.2; projected norm 1.06×10^{-4}) — while the spurious mode, being genuinely solenoidal (divergence ratio $\approx 6 \times 10^{-15}$), *survives*; (3) a port-aware rank-1 re-admission of the junction flux direction yields a projector that retains all six physical modes at $\leq 0.029\%$ worst case versus Palace. What the artifact *is*: under the L-doubling harness it moves $3.4528 \to 2.4449$ GHz (ratio 0.7081, the junction $1/\sqrt{L}$ law) with 99.4% of its stiffness energy in the K_{port} surface term — a second LC-like resonance of the port-surface operator S_Γ, which Palace's `LumpedPort` elimination represents differently and therefore never sees. The headline comparison deliberately remains the committed ungauged benchmark (the port-aware solver shipped after the headline measurement); the filter now excludes a *characterized* artifact rather than an unexplained one. Figure 4 contrasts the spectra. The arc measures precisely what Palace's divergence-free projection buys in a production eigensolver, and what a lumped-element formulation demands beyond it.

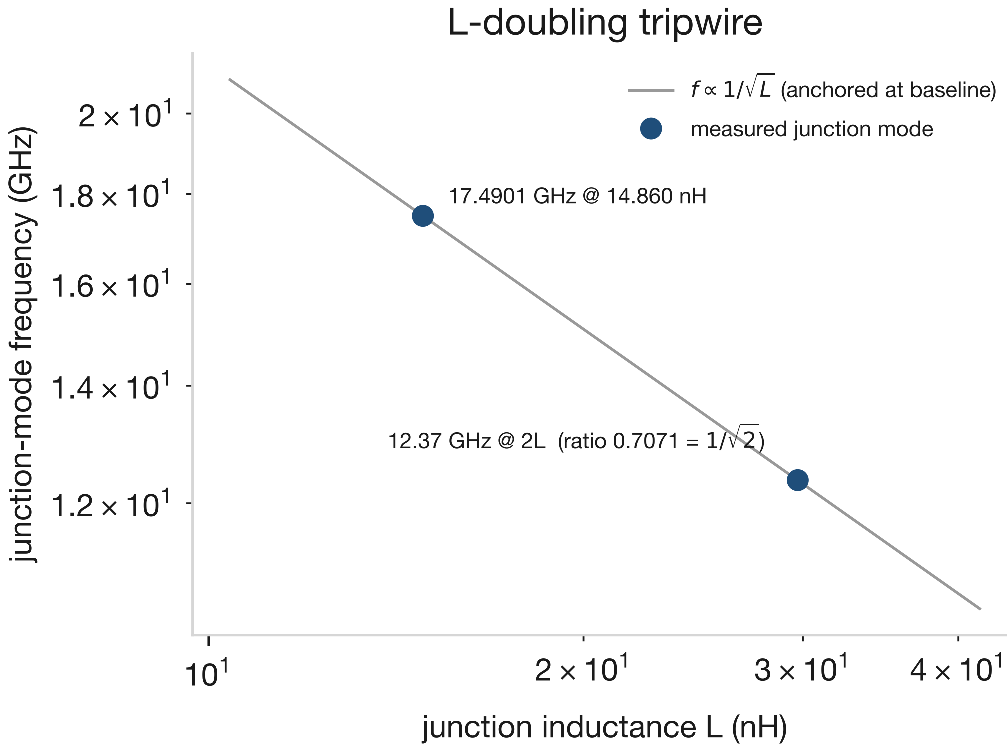

Figure 3: Josephson inductance-scaling tripwire: junction-mode frequency versus junction inductance L on log-log axes. Doubling L moves the junction mode from 17.4901 to 12.37 GHz — committed ratio $0.7071 = 1/\sqrt{2}$ — exactly as $f \propto 1/\sqrt{L}$ requires. Measured values from the committed `benchmarks/transmon_eigen/results.toml` tripwire block.

7 The Forward LOM Chain: From Fields to E_C

The lumped-oscillator-model (LOM) branch computes the transmon's Hamiltonian parameters from the *electrostatic* problem on the same committed mesh — and it is the chain the remainder of this paper differentiates. The mesh's metal is split into three node-disjoint conductors (ground+resonator, feedline, island); a tensor-ε electrostatic solve yields the Maxwell capacitance matrix by the energy-reaction form $C_{ij} = \phi_i^{\mathsf{T}} K \phi_j$; the island self-capacitance is reduced for the floating feedline ($C_\Sigma = C_{ii} - C_{if}^2/C_{ff}$); and the charging energy follows as $E_C/h = e^2/(2C_\Sigma h)$, with the transmon spectrum evaluated charge-basis-exactly after Koch et al. (2007).

Measured on the committed fixture (`benchmarks/transmon_quantum/results.toml`): $C_\Sigma = 136.7\,\mathrm{fF}$, hence $E_C = 0.142\,\mathrm{GHz}$, $E_J/E_C = 77.6$ (with E_J from the junction inductance), $\omega_{01} = 3.38\,\mathrm{GHz}$, and anharmonicity $\alpha = -0.158\,\mathrm{GHz}$ — a plausible transmon computed end-to-end from committed geometry, with the $\alpha \approx -E_C$ sanity holding.

The honest negative that travels with this chain. The extracted $C_\Sigma = 136.7\,\mathrm{fF}$ *exceeds* the $\sim$90 fF design-anchor value back-solved from the blog workflow's expected $E_C = 0.2156\,\mathrm{GHz}$. The committed benchmark diagnoses this as a model-definition difference, not a truncation artifact (grounding the exterior wall moves C_Σ by a relative 6×10^{-5}; the scalar-versus-tensor sapphire treatment shifts it 0.75%). We report the extracted numbers and the discrepancy; we do not retrofit to the anchor. Section 9 returns to this gap as an optimization target — and reports honestly how

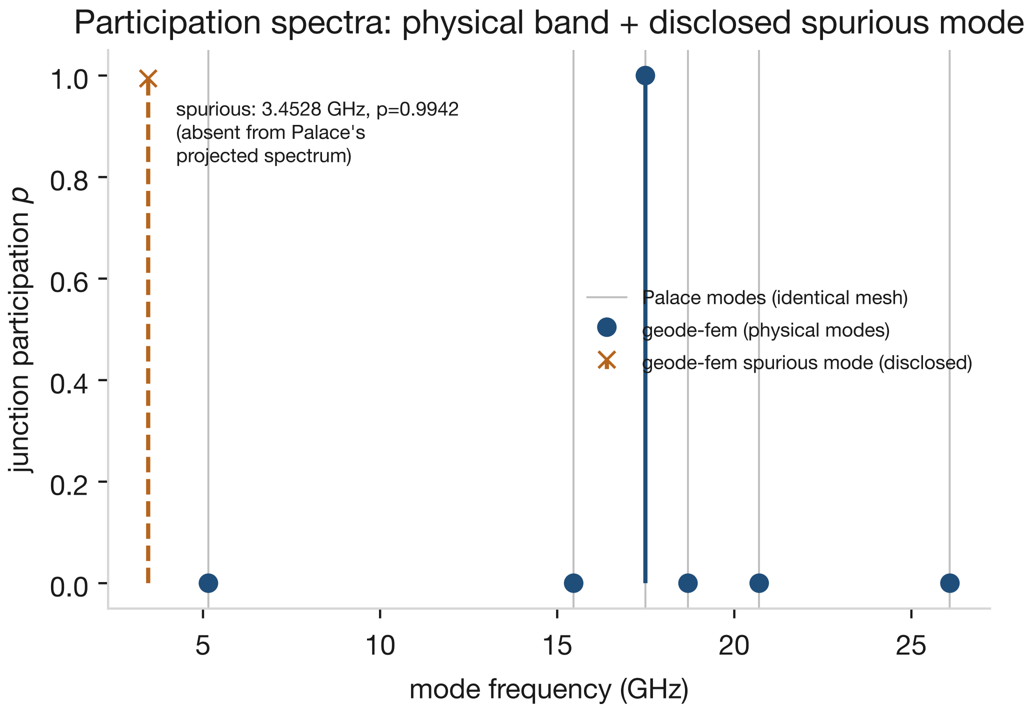

Figure 4: geode-fem junction-participation spectrum on the identical mesh, with Palace's mode locations (from the committed `eig.csv`) overlaid. The physical junction LC mode appears at 17.49 GHz in both solvers' spectra (geode-fem stiffness participation $p = 1.000$); geode-fem's un-projected headline Lanczos additionally leaks one spurious junction-localized mode at 3.4528 GHz ($p = 0.9942$) with no Palace counterpart — characterized in Section 6.2 as a port-surface artifact obeying the junction $1/\sqrt{L}$ law. Palace's field-based port-EPR is a differently normalized diagnostic and is not plotted on geode-fem's participation axis (Section 5).

far a fixed-topology parametrization gets toward closing it.

8 Differentiable Sensitivities: The Discrete-Adjoint Layer

8.1 The 2×2 sensitivity matrix

For a solve $A(\varepsilon, X)\, x = b(X)$ and a scalar observable $g(x)$, the discrete adjoint $A^\mathsf{T}\lambda = \partial g/\partial x$ turns the parameter gradient into explicit contractions:

$$\frac{dg}{d\varepsilon_k} = -\lambda^\mathsf{T}\frac{\partial A}{\partial \varepsilon_k}x, \qquad \frac{dg}{dX} = \lambda^\mathsf{T}\frac{\partial b}{\partial X} - \lambda^\mathsf{T}\frac{\partial A}{\partial X}x \tag{4}$$

(real case; the complex driven case takes $2\,\mathrm{Re}[\cdot]$ of the same contractions with the *transpose*, not conjugate-transpose, solve — the pencil is complex-symmetric, so the adjoint solve reuses the forward LU factorization). Everything on the right-hand side is assembly-level and therefore differentiable on the tensor substrate: the element Jacobians $\partial A/\partial X$ and $\partial b/\partial X$ are *exact*, computed by forward-mode dual numbers through the same closed-form element kernels the solver assembles — no finite-difference truncation enters the gradient. Each gradient costs one forward plus one adjoint solve sharing a single factorization (asserted programmatically: `n_factorizations` $= 1$).

Four adjoint modules are merged and validated, spanning material and geometry sensitivities on both operators (Table 2). Each carries a committed test that the analytic gradient match a central finite difference *of the full independent pipeline* (perturb the parameter, re-assemble, re-solve

Table 2: The merged 2×2 discrete-adjoint sensitivity matrix. Each cell: one forward + one adjoint solve sharing a single factorization; element Jacobians exact (forward-mode duals through the production kernels); validated against central finite differences of the full pipeline, with mutation tests. "FD rel-err" is the observed agreement of the analytic gradient with the FD reference in the merged test suite (test acceptance bounds are looser; see the module sources). Modules in `crates/geode-core/src/`.

Operator	Parameter	Module (issue / PR)	FD rel-err
scalar electrostatic (SPD)	material $\partial/\partial\varepsilon$	`adjoint` (#570 / #573)	$\sim 3 \times 10^{-8}$
scalar electrostatic (P1)	geometry $\partial/\partial X$	`shape` (#571 / #575)	$\sim 1 \times 10^{-9}$
driven H(curl)	material $\partial/\partial\varepsilon$	`driven::adjoint` (#576 / #579)	2.3×10^{-5} (worst region)
driven H(curl)	geometry $\partial/\partial X$	`driven::shape` (#577 / #581)	2.2×10^{-9}, 1.2×10^{-8} (two maps)

through the public solver path, recompute g) — and, load-bearingly, mutation tests proving the tolerances bite: the classic complex-adjoint conjugation error (A^{H} for A^{T}) is rejected by the FD check, as is a dropped term. The driven-shape case surfaced a physics subtlety worth recording: the current-source right-hand side $b = \mathrm{i}\omega\mu_0 \int N \cdot J$ is geometry-dependent, so the shape gradient carries a $\partial b/\partial X$ term absent from the material case — omitting it fails the finite-difference cross-check outright.

The matrix is deliberately 2×2: material and geometry are the two design-parameter classes, and the scalar and driven operators are the electrostatic (LOM) and RF (S-parameter/EPR-workhorse) branches of the solver. The LOM optimization of Section 9 exercises the scalar column end-to-end; the driven column is merged and validated at the operator level and composes with the driven solve that geode-fem's GPU path accelerates — RF shape optimization is the natural follow-on, not exercised here.

8.2 The capacitance→ E_C chain, and a vanishing adjoint

Composing the shape adjoint with the LOM chain of Section 7 gives $\partial(C, E_C, \alpha)/\partial(\text{geometry})$ (merged as `shape::capacitance_shape_gradient`, #583 / PR #586). The composition carries an elegant simplification. The capacitance observable is the stored field energy, $C = 2W = \phi^{\mathsf{T}} K(X)\phi$, and W is *variationally stationary* in the solved potential ϕ: at the forward solution the free rows of the cotangent $K\phi$ vanish (equilibrium), so the adjoint solution $\lambda \approx 0$ and the entire shape derivative collapses to the explicit-geometry term

$$\frac{dC}{dX} \;=\; \phi^{\mathsf{T}} \, \frac{\partial K}{\partial X} \, \phi \qquad \text{(at the solution)}, \tag{5}$$

a Hellmann–Feynman-like result: no adjoint back-solve is needed for the capacitance itself. The implementation keeps the general adjoint form (the mutation test measures the implicit part at $< 10^{-8}$ of the total, confirming the collapse numerically rather than assuming it), and chains through $\partial E_C/\partial C_\Sigma = -e^2/(2C_\Sigma^2 h)$ exactly. Validation is double: against the analytic parallel-plate law ($\partial C/\partial d = -\varepsilon_0\varepsilon_r$ per unit area under a gap scaling; $\partial C/\partial A = +2\varepsilon_0\varepsilon_r$ under an area scaling — opposite signs, both matched) and against central finite differences of the full pipeline, with observed relative errors at the 10^{-9} level.

9 Gradient-Based Optimization: Demonstrations and an Honest Negative

Two demonstrations, in deliberately labeled honesty tiers: a closed-loop proof on an analytic fixture, then the real device mesh. Both use damped Newton on $E_C(\theta) - E_C^{\text{target}}$ (equivalently $C(\theta)$) with the analytic $dE_C/d\theta$ from Section 8.2; each Newton step costs one forward plus one adjoint solve on a single factorization. The framing claim is capability and step-count, not wall clock: a derivative-free

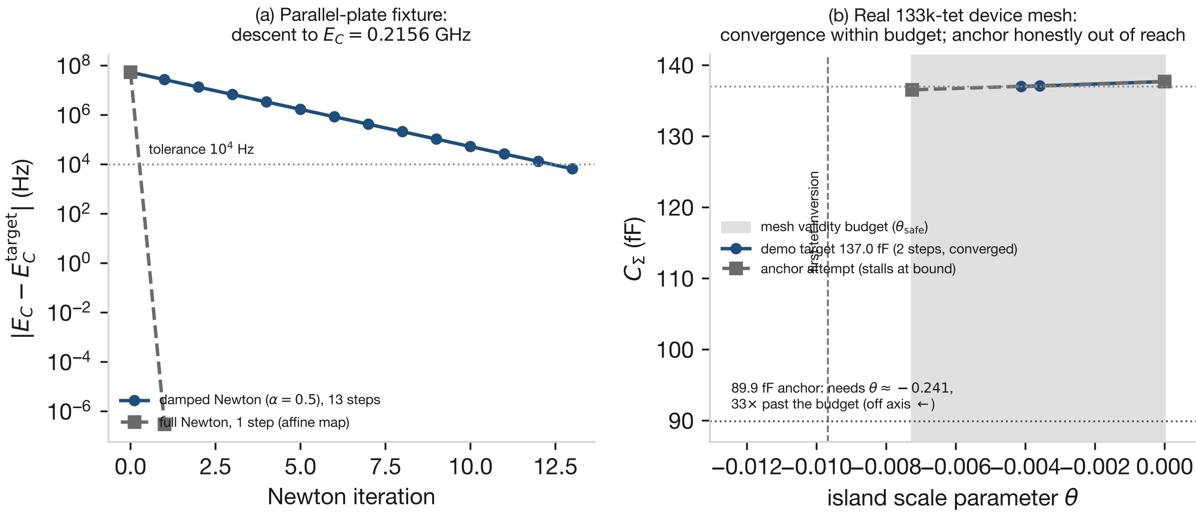

Figure 5: Gradient-based optimization to a target E_C via the discrete-adjoint shape gradient. (a) Parallel-plate loop proof: $|E_C - E_C^{\mathrm{target}}|$ versus Newton iteration; full Newton converges in one step (the parametrization is affine — expected, and disclosed), the damped $\alpha = 0.5$ run supplies the 13-step descent curve; the converged design is confirmed by an independent fresh forward solve at relative error 1.4×10^{-15} (`benchmarks/transmon_diffopt/results.toml`). (b) The real 133k-tet DeviceLayout mesh under the fixed-topology island-scale parameter θ: a within-budget demonstration target ($C_\Sigma = 137.0\,\mathrm{fF}$) converges in two genuine (non-affine) Newton steps, fresh-solve confirmed at 5.6×10^{-6}; the bounded run toward the 89.9 fF design anchor stalls honestly at the mesh validity budget $\theta_{\mathrm{safe}} = -0.0073$ (first tetrahedron inversion at $\theta = -0.0097$), 33× short of the $\theta \approx -0.241$ the anchor requires (`benchmarks/transmon_diffopt/pad_results.toml`).

loop (the parameter sweeps of current practice) spends N_{params} extra forward solves per step to estimate what the adjoint delivers exactly for one. No timing comparison against HFSS/Palace-based optimizers is measured or claimed.

9.1 Loop proof: Newton to a target E_C on a parallel-plate fixture

On a parallel-plate capacitor fixture ($\varepsilon_r = 3$; 343 nodes, 1,296 tets; θ scales the gap, so $d = L(1+\theta)$), Newton iteration drives the design from $C_0 = 120.0\,\mathrm{fF}$ ($E_C = 0.1614\,\mathrm{GHz}$) to the design-anchor target $E_C = 0.2156\,\mathrm{GHz}$. Full Newton converges in *one* step — and we state the reason rather than dress it up: $C \propto 1/(1+\theta)$ makes $E_C(\theta)$ *affine* in θ, so one-step convergence is expected, and this tier proves the loop end-to-end (gradient, chain rule, update, convergence test), not a hard optimization. The converged $\theta = 0.335657909$ matches the closed-form optimum to 3×10^{-15}. The honesty check that matters: a *fresh, independent* forward solve at the converged geometry — assemble, solve, extract, no adjoint machinery — reproduces the target E_C to a relative error of 1.4×10^{-15}. A damped run ($\alpha = 0.5$) converges in 13 steps and supplies the descent illustration (Figure 5a). At the hit target, the Koch-exact spectrum gives $E_J/E_C = 51.0$, $\omega_{01} = 4.128\,\mathrm{GHz}$, $\alpha = -0.247\,\mathrm{GHz}$ — the $\sim$4 GHz qubit regime the design anchor implies. All trajectory values are committed in `benchmarks/transmon_diffopt/results.toml`.

9.2 The real device: a validated gradient, a genuine convergence, and a 33× shortfall

The second tier runs on the committed 133k-tet SingleTransmon mesh itself, with a fixed-topology shape parameter θ that scales the island conductor in-plane about its centroid (101 island nodes move; the mesh connectivity is fixed). Three results, in descending order of comfort (`benchmarks/transmon_diffopt/pad_results.toml`):

The gradient is real on the real mesh. At $\theta = 0$ the adjoint gradient $\partial C_\Sigma/\partial\theta = 198.198\,\text{fF}$ is validated against a central finite difference of the *full independent pipeline* (move the island nodes, re-assemble, re-solve, re-extract): relative error 1.15×10^{-4} at $h = 10^{-4}$, with a clean $O(h^2)$ truncation sweep ($1.2 \times 10^{-2} \to 1.0 \times 10^{-3} \to 1.15 \times 10^{-4} \to 1.0 \times 10^{-5}$ as h falls $10^{-3} \to 3 \times 10^{-5}$). This is the paper's load-bearing real-geometry validation: the same machinery that passed at 10^{-9}-level on analytic fixtures holds up on production geometry.

The loop genuinely converges on the real mesh. A demonstration target within the mesh's validity budget ($C_\Sigma = 137.0\,\text{fF}$) converges in two Newton steps — genuinely non-affine now (the gradient changes by $\sim$21% along the path) — and the converged design is confirmed by a fresh, independent multi-conductor extraction at relative error 5.6×10^{-6} (Figure 5b). The entire real-device run (validation sweep plus both optimization runs) takes 10.7 s.

The honest negative: the anchor is out of this parametrization's reach. The natural target is the $\sim$90 fF anchor of Section 7 (89.9 fF; $E_C = 0.2155\,\text{GHz}$ — the last digit differs from the 0.2156 GHz quoted elsewhere because this run targets the rounded 89.9 fF rather than the back-solved 89.843 fF; both values are artifact-faithful). The $\theta = 0$ gradient puts it at $\theta \approx -0.241$ — but the fixed-topology map inverts its first tetrahedron at $\theta = -0.0097$, and the run's conservative budget (minimum tet-volume ratio 0.25) caps $\theta_{\text{safe}} = -0.0073$, where the bounded Newton run stalls at $C_\Sigma = 136.5\,\text{fF}$: a 33× shortfall. The diagnosis is geometric, not a gradient failure: the island's junction-attachment nodes sit $\sim$225 μm from the island centroid, so the centroid scale drags them against $\sim$0.7 μm junction-region tetrahedra. Closing the anchor gap is therefore a *mesh-morphing and parametrization* problem — a smoothed velocity field or per-step remeshing, multi-parameter pad/gap/lead controls that remove the junction lever arm, and a tensor-ε extension of the shape chain (the committed run uses the scalar trace-averaged sapphire; the scalar-versus-tensor difference is quantified at 0.75% in the artifact) — all named follow-ons, none of which the present paper claims. We regard this shortfall as a result, not an embarrassment: it is precisely the boundary information a design-gradient capability must publish before anyone builds an optimizer on it.

10 Roadmap: Differentiating the Eigenmode Branch

The LOM branch above differentiates an *electrostatic* solve. The transmon's dynamical figures-of-merit — resonator frequency, participation-ratio and EPR self/cross-Kerr (Minev et al., 2021a; Nigg et al., 2012) — live in the *eigenmode* problem, and that branch is **not differentiable yet** in geode-fem. The paper commits to the explicit split:

Branch	Observables	Differentiable?
LOM (electrostatic)	E_C, $\alpha \approx -E_C$, coupling C	**yes** (Sections 8–9)
eigenmode / EPR	resonator f, participation, Kerr	**not yet** (this section)

Differentiating an eigenpair needs the adjoint-eigenpair (Hellmann–Feynman / Nelson) formulas (Nelson, 1976), which require solving the interior eigenproblem robustly at the physical shift — and geode-fem's matrix-free interior eigensolve is blocked at exactly that shift. The blocker is measured and committed (`benchmarks/transmon_bench_cpu/sigma4p5_deepshift_characterization.md`): at $\sigma = 4.5\,\text{GHz}$ on the 133k-DOF fixture, a *single* inner AMS-preconditioned MINRES solve (outer Lanczos step 0) never converges — it plateaus on a nearly flat residual tail below $\sim10^{-5}$ and was still running at 14,300 inner iterations ($\|r\| \approx 2.6 \times 10^{-6}$) when killed. The plateau is *coarse-solve-invariant* — an exact coarse solve stalls the same way — so the limiter is the SPD-proxy preconditioner $K + |\sigma|M$ itself, not the coarse solve. Nor is there a single inner tolerance that is both cheap and correct: at an inner tolerance of 10^{-2} the entire eigensolve completes in ∼56 s but returns the non-physical $\lambda \approx 0$ gradient near-kernel cluster (0.42–0.64 GHz), not the physical band. The named path — not built, cited as method, not result — is a preconditioned Helmholtz-projected Jacobi–Davidson iteration, applying the gradient-kernel projection inside the correction equation, for which Liang et al. (2026) provide the current adaptive-multilevel treatment of exactly this singular Maxwell eigenproblem; combined with the Nelson/Hellmann–Feynman eigenpair derivatives, that is the route to differentiable eigenmode/EPR quantities. Until it lands, no claim in this paper extends to derivatives of the eigenmode spectrum.

11 Performance Context

Performance is context here, not contribution: on the measured record there is no corner where geode-fem beats Palace at scale, and the capability claim of this paper does not depend on one. We report the matched CPU cell (with the corrected large-scale finding) and the GPU cell (correctness proven; scaling an honest negative) for completeness and because the honest accounting bounds what the differentiable solver costs to run.

11.1 CPU cell: a matched head-to-head, and a flop-and-fill crossover

All CPU measurements ran on a single EC2 `m6i.4xlarge` (8 physical cores / 16 vCPUs, ∼61 GB usable) in `us-west-2`.[3] The cell is matched: same box, same session, identical mesh and eigenproblem, both solvers at 1 and 8 cores/ranks, three workloads (physical target: 6 modes at 4.5 GHz; off-target: 12 modes at 20 GHz; uniformly refined ∼1.16M-DOF mesh), full pipeline, mean of $n = 3$ runs (`benchmarks/transmon_bench_cpu/results.toml`).

Three readings, none of them a parallelization claim. *Per-core efficiency:* at the physical target, geode-fem's serial direct factorization on one core (28.7 s) beats Palace on eight ranks (44.5 s) in absolute wall clock at ∼12× fewer core-seconds (28.7 versus 356.0); geode-fem's own 8-thread run gives essentially no speedup (29.0 s; tracked as issue #518), so the advantage is direct-versus-iterative throughput, not parallelism. *Target robustness:* the direct shift-invert is target-insensitive while Palace's iterative solve degrades off-target, so the gap *widens* at 20 GHz — 6.7× serial (248.0/36.8), 2.4× at 8-wide (64.7/26.6). *The corrected large-scale finding:* the direct factorization trades memory and super-linear fill/flops for that speed and robustness, and the trade stops paying below 1M DOFs. On the ∼1.16M-DOF mesh, given adequate memory (a 128 GB box), the direct solve *does complete* — setup 35.78 s, solve 529.75 s, total 565.5 s at 92.2 GB peak RSS — but it *loses to Palace on both axes*: Palace finishes the same mesh in 423.12 s at 4.1 GB/rank (∼33 GB aggregate). An earlier "never completes" reading (the OOM kill at 63.9 GB on the ∼61 GB box) was a small-box truncation artifact and is retracted in favor of this measurement: the crossover is a *flop-and-fill deficit, not merely a memory wall* — even with abundant memory, the direct path is a small-to-medium-scale win that loses at scale. The architecture trade-off, stated symmetrically: geode-fem wins small-to-medium on speed, per-core efficiency, and target robustness; Palace wins at scale on both wall clock and memory. (Palace at 16 ranks on hyperthreaded vCPUs was refused by MPI process binding; we exclude it rather than report a degraded configuration.) The once-planned matrix-free interior eigensolve is no longer offered as the scale answer: its deep-shift inner solve is the measured wall of Section 10, and the honest scale story for the direct path is the one in Table 3.

[3] On-demand cost approximately $0.77/hour at the time of the runs; the CPU cell is reproducible for a few dollars.

Table 3: Matched CPU-cell wall-clock and memory, full pipeline, `m6i.4xlarge`, same box and session (mean of $n = 3$). geode-fem `@3174015`: serial/threaded faer sparse-LU shift-invert Lanczos (direct); Palace `@fba6a5b`: distributed MPI Krylov + AMS (iterative). Palace memory is per-rank maximum RSS, not an aggregate. At the physical target geode-fem on *one* core beats Palace on *eight* ranks in absolute wall clock; at ∼1.16M DOFs the direct solve *completes given adequate memory but loses to Palace on both wall clock and memory* — the crossover is a flop-and-fill deficit, not merely a memory wall. Values from the committed `benchmarks/transmon_bench_cpu/results.toml` (`[matched.*]` tables) and, for the 128 GB-box completion row, the committed run log `benchmarks/transmon_bench_cpu/geode_runs_1p16M_2026-07-15.log`.

Workload	Solver (cores)	Wall (s)	Peak RSS
Physical target: 6 modes @ 4.5 GHz, 133,108 interior DOFs			
	geode-fem (1)	28.7	3.1 GB
	geode-fem (8)	29.0	3.1 GB
	Palace (1)	130.9	0.5 GB/rank
	Palace (8)	44.5	0.5 GB/rank
Off-target: 12 modes @ 20 GHz, 133,108 interior DOFs			
	geode-fem (1)	36.8	3.1 GB‡
	geode-fem (8)	26.6	3.1 GB‡
	Palace (1)	248.0	0.5 GB/rank‡
	Palace (8)	64.7	0.5 GB/rank‡
Large scale: 6 modes @ 4.5 GHz, 1,157,564 interior DOFs			
	geode-fem (1), ∼61 GB box†	killed at ceiling	63.9 GB (truncated)
	geode-fem (1), 128 GB box†	565.5	92.2 GB
	Palace (8)	423.12	4.1 GB/rank (∼33 GB aggr.)

†The geode-fem large-scale rows depart from the same-box protocol of the matched cell and are marked accordingly: the ∼61 GB-box run was SIGKILL-ed mid-factorization at the memory ceiling (its 63.9 GB "peak" is a truncation of the true footprint, not a measurement of it), and the completion row was re-run on a 128 GB `r6i`-class box (committed log above), where the true peak is 92.2 GB. Palace's row is from the original matched session.

‡The committed `[matched.off_target.*]` tables record wall times only; the off-target Peak RSS cells repeat the physical-target measurements of the same session and the same 133,108-DOF pencil, and are not independently recorded in the artifact.

11.2 GPU cell: correctness proven, scaling an honest negative

geode-fem's GPU path accelerates the *driven* solve (matrix-free apply + on-device COCG), not the eigensolve of the credential, and all GPU execution is CUDA-f32 (Section 3). On 2026-07-14, geode-fem code executed on a physical GPU for the first time (EC2 `g6e.xlarge`, 1× NVIDIA L40S 46 GB, driver 595.71.05, CUDA 13.2): the two CUDA-f32 correctness smokes — the matrix-free Nédélec matvec and the on-device COCG — pass against the f64 CPU reference within f32-appropriate tolerances (15.0 and 3.3 s wall, durations dominated by GPU init and kernel JIT — correctness evidence, not performance).

The scaling study (committed as `benchmarks/gpu_driven_scaling/results.toml`; a σ-lossy lumped-port cube fixture at fixed frequency, mesh-refined from 1,854 to 25,695 edges, all sixteen cells on the same host) is an honest negative: **GPU-f32 matrix-free loses to every CPU configuration at every measured size** (Table 4). Computed from the artifact's unrounded medians, the assembled-CSR COCG on CPU is 136× faster at 1,854 edges (displayed: 0.032 versus 4.39 s) and still 44× faster at 25,695 (1.86 versus 81.8 s); even the direct LU is 13.5× faster at the top size (6.04 versus 81.8 s). The mechanism is kernel-launch domination (∼28 ms per iteration at the top size versus ∼0.63 ms for assembled-CSR on CPU) — 25.7k edges is far too small to fill

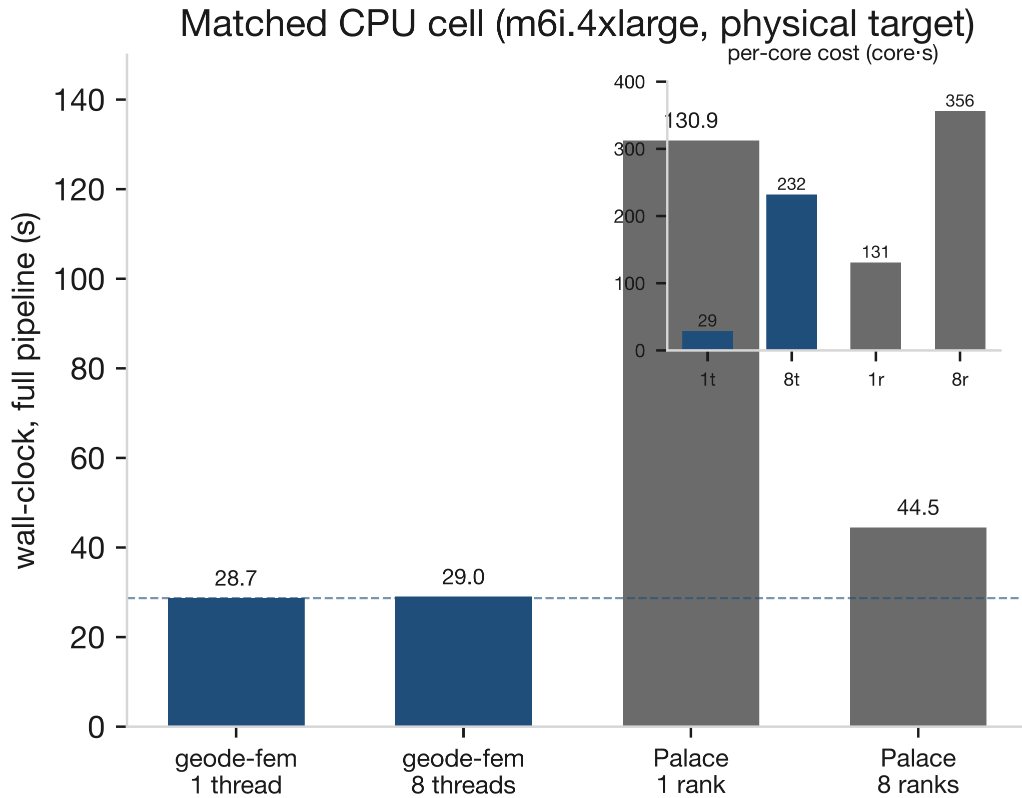

Figure 6: Matched CPU-cell wall-clock comparison at the physical target (6 modes, 4.5 GHz, 133k DOFs): geode-fem at 1 and 8 threads versus Palace at 1 and 8 MPI ranks. The inset reports per-core cost as core-seconds — geode-fem on one core consumes 28.7 core-s against Palace's 356.0 at 8 ranks ($\sim$12×). Full-pipeline timings from Table 3.

an L40S — and f32 accuracy compounds the verdict (relative error degrading to 1.2×10^{-3} with refinement; nondeterministic convergence at the top size). One directional positive survives, and it is the one the architecture predicts: against the *same matrix-free algorithm on CPU*, the GPU's deficit falls monotonically to parity at $\sim$26k edges (81.8 versus 82.1 s). We state the GPU position as trajectory, not achievement, with tracked levers (kernel fusion; on-device f64/mixed-precision when cubecl ships it; preconditioning) — and we prefer publishing the bound to implying the win. This cell is a cube fixture family, not the transmon eigenproblem, and its CPU baselines (4-vCPU host) are not comparable to the `m6i` numbers of Table 3.

12 Reproducibility

Every artifact behind every number in this paper is committed and scripted.[4]

- **Geometry and mesh.** Generated by DeviceLayout.jl v1.15.0's `SingleTransmon` example (AWS Center for Quantum Computing, 2025); the fixture generator is committed, and the MSH 4.1 fixture is pinned by SHA-256 (`5b3ff4c3...b33dd`; full hash and provenance in the committed `transmon_smoke.provenance.txt`).

[4]Repository URL and archival DOI for the geode-fem source to be added at submission. TODO(operator).

Table 4: Driven-solve wall-clock scaling, solve-only medians of $n = 3$ timed repetitions (warm-up excluded), same host (`g6e.xlarge`: 4 vCPU + 1× L40S). One fixture (σ-lossy lumped-port cube) at fixed ω, mesh-refined across the columns. f64 iterative configurations at tolerance 10^{-8}; the f32 GPU configuration at 10^{-6} (its recurrence floor sits above 10^{-8}). The accuracy row is the full-field relative L_2 error of the GPU-f32 solution against the Direct-f64 reference. At 25,695 edges the GPU-f32 configuration converged in the committed single-frequency runs but not in its five-point frequency sweep, and run-to-run convergence is nondeterministic at that size (f32 floor). Values verbatim from the committed `benchmarks/gpu_driven_scaling/results.toml`.

	Mesh size (Nédélec edges)			
Configuration	1,854	5,859	13,428	25,695
Direct faer LU (CPU f64), s	0.024	0.203	1.540	6.036
Assembled-CSR COCG (CPU f64), s	0.032	0.198	0.709	1.865
Matrix-free COCG (CPU f64), s	1.652	8.885	30.234	82.056
Matrix-free COCG (GPU f32), s	4.388	13.351	34.381	81.764
GPU-f32 accuracy vs. Direct-f64 (rel. L_2)	1.26×10^{-4}	2.93×10^{-4}	4.60×10^{-4}	1.20×10^{-3}

- **The MSH conversion gotcha.** MFEM's MSH 4.1 reader rejects the DeviceLayout multi-entity-block file, so the Palace runs consume an MSH 2.2 conversion (`gmsh -save -format msh2`). The conversion is node-preserving (all 22,684 nodes unchanged, physical groups preserved) and physics-neutral: eigenvalues reproduce bit-for-bit across the conversion. Documented in the committed `palace_config.provenance.txt` because it is exactly the kind of silent pipeline step a same-mesh claim must disclose.

- **Credential artifacts.** The Palace configuration and provenance under `reference/fixtures/transmon_palace/`; geode-fem's benchmark definitions, the agreement numbers, the L-doubling tripwire (17.4901 → 12.37 GHz, committed ratio 0.7071), the spurious-mode measurements (3.4528 GHz, $p = 0.9942$; L-doubling ratio 0.7081), and the full gauge/projection arc of Section 6.2 (tree–cotree counts and the 1.64% shift; bulk-projection divergence ratio 50.2 and projected norm 1.06×10^{-4}; port-aware acceptance at $\leq 0.029\%$) in `benchmarks/transmon_eigen/results.toml`, each pinned by a release-tier regression test. Palace's raw outputs (`eig.csv`, per-port participation, run log) are committed; a rerun reproduced `eig.csv` bit-for-bit.

- **Precision of the committed agreement numbers.** `benchmarks/transmon_eigen/results.toml` stores geode-fem frequencies at three decimals while the Palace column carries full precision, so the committed per-mode `rel_err_pct` is computed against the rounded geode-fem values; at full precision the agreement is slightly better (approximately 0.029% worst case). We quote the committed-artifact numbers throughout and disclose the rounding here. Derived ratios quoted in the text (core-second and speedup multipliers) are computed from the displayed table values unless marked as committed.

- **LOM chain.** The three-conductor capacitance extraction and Koch-exact spectrum ($C_\Sigma = 136.7\,\mathrm{fF}$, $E_C = 0.142\,\mathrm{GHz}$, and the anchor-gap diagnosis of Section 7) in `benchmarks/transmon_quantum/results.toml`; pinned by the release test `crates/geode-core/tests/transmon_quantum.rs`.

- **Adjoint layer.** The four sensitivity modules with their FD-and-mutation test suites: `crates/geode-core/src/adjoint.rs` (PR #573); `crates/geode-core/src/shape.rs` (PRs #575/#586; this module also carries `capacitance_shape_gradient`); `crates/geode-`

core/src/driven/adjoint.rs (PR #579); and crates/geode-core/src/driven/shape.rs (PR #581).

- **Optimization demonstrations.** The parallel-plate loop proof (full trajectories, closed-form check, fresh-solve confirmation) in benchmarks/transmon_diffopt/results.toml; the real-device FD validation, demonstration convergence, and the bounded anchor attempt with its mesh-safety budget in benchmarks/transmon_diffopt/pad_results.toml; both consumed by crates/geode-core/tests/transmon_diffopt.rs.

- **Performance artifacts.** The matched CPU cell in benchmarks/transmon_bench_cpu/results.toml; the 1.16M-DOF completion run log at benchmarks/transmon_bench_cpu/geode_runs_1p16M_2026-07-15.log; the GPU scaling cell (all sixteen cells, iteration counts, recomputed true residuals, DNF handling, same-host provenance) in benchmarks/gpu_driven_scaling/results.toml; the deep-shift eigensolve characterization at benchmarks/transmon_bench_cpu/sigma4p5_deepshift_characterization.md.

- **Hardware disclosure.** CPU cell: EC2 m6i.4xlarge (8 physical cores / 16 vCPU, 64 GB), us-west-2; the 1.16M-DOF completion row: a 128 GB r6i-class box (see the committed log). GPU cell: EC2 g6e.xlarge (4 vCPU, 30 GB; 1× NVIDIA L40S 46 GB; driver 595.71.05, CUDA 13.2).

- **Versions.** geode-fem commit 3174015 (headline credential and CPU cells; the adjoint/diffopt layers are merged post-3174015 and cited by PR number); Palace commit fba6a5b; DeviceLayout.jl v1.15.0; Burn/burn-cuda 0.21.

13 Discussion

What the credential does and does not establish. In V&V terms (Oberkampf and Roy, 2010; Roache, 1997) the cross-validation is code verification and code-to-code comparison, not validation against nature: two independent implementations agreeing to $\leq 0.033\%$ on the identical discrete problem, with analytic tripwires pinning the junction physics. It does not certify that either solver matches a measured device — that is what Ye et al. (2025)-style experimental benchmarks address — but it rules out the large error class living in formulation, assembly, boundary handling, and eigensolver machinery. For this paper's purpose it establishes the specific premise the gradients need: the forward solutions being differentiated are production-accurate.

What the differentiable-design evidence supports. The proven capability is bounded and we state its boundary. Proven: the 2×2 adjoint matrix at operator level (FD-validated with mutation tests); the capacitance$\rightarrow E_C$ chain with its stationarity collapse; a closed Newton loop confirmed by independent fresh solves on both an analytic fixture and the real 133k-tet mesh; and a real-mesh gradient validated at 1.15×10^{-4} against the full independent pipeline. Not proven, and not claimed: eigenmode/EPR derivatives (the Section 10 wall); anchor-reaching real-device optimization (the 33× parametrization shortfall of Section 9.2); tensor-ε shape chains; multi-parameter optimization at scale; and any wall-clock superiority over sweep-based workflows. The capability is also not qubit-specific — shape and material sensitivities of electrostatic and driven-EM observables apply to RF and photonic device design generally; the transmon is the demonstration vehicle with the clearest documented unmet need.

Threats to validity. A common-mode error — both solvers wrong the same way — cannot be excluded by cross-comparison alone; the analytic anchors and disjoint implementations make it unlikely for the junction physics. The optimization demonstrations are single-parameter by design; nothing here demonstrates multi-parameter or constrained optimization behavior. The parallel-plate tier's one-step convergence is an affine-map property, disclosed as such. The real-device chain uses the scalar trace-averaged sapphire (tensor-ε delta quantified at 0.75%). Performance numbers are $n = 3$ measurements on one instance type per cell and support architecture-level readings, not microsecond claims; the large-scale completion row is a different (128 GB) box, marked as such in Table 3.

Limitations. (i) One geometry, one mesh, first order only — higher-order and mesh-refinement convergence is future work (a $p = 2$ Palace configuration is committed for that purpose). (ii) The lossless model leaves readout ports open; resistive terminations (hence complex-symmetric pencils and mode Qs) are out of scope. (iii) The credential eigenpath remains the ungauged committed benchmark; the port-aware projection retains all six modes at $\leq 0.029\%$ but is a composite solver, and the solenoidal port artifact is characterized and filtered rather than eliminated. (iv) The CPU per-core win is small-to-medium-scale only: at $\sim$1.16M DOFs the direct solve completes given memory but loses to the distributed iterative reference on both wall clock and memory — a flop-and-fill crossover below 1M DOFs, per the committed 128 GB-box measurement. (v) GPU evidence covers driven-solve correctness and an honest scaling negative in f32 on a cube fixture family; no GPU eigensolve path exists. (vi) The differentiable branch is electrostatic/LOM only (eigenmode/EPR is roadmap), currently scalar-ε for the shape chain, and its real-device demonstration is bounded by the fixed-topology parametrization budget.

14 Conclusion

Superconducting-qubit electromagnetic design iterates a solver that cannot say which way to move. This paper demonstrated the missing derivative: a discrete-adjoint layer over a tensor-native FEM solver that turns one extra solve per step into exact, solver-derived sensitivities — a validated 2×2 matrix of material and geometry gradients on the electrostatic and driven $H(\mathrm{curl})$ operators — composed into a differentiable capacitance$\rightarrow E_C$ chain whose shape derivative collapses, by variational stationarity, to a Hellmann–Feynman-like explicit term. The capability was exercised honestly at two tiers: a closed Newton loop that hits a 0.2156 GHz target E_C and survives an independent fresh-solve check at 1.4×10^{-15} (one-step convergence disclosed as the affine-map property it is), and the real 133k-tet device mesh, where the gradient is FD-validated at 1.15×10^{-4}, a within-budget target converges in two genuine Newton steps — and the $\sim$90 fF design anchor sits 33$\times$ beyond the fixed-topology parametrization's mesh budget, a mesh-morphing problem we publish as the capability's current boundary. The claim is held to a cross-validation evidence standard: the same solver reproduces Palace to $\leq 0.033\%$ across all six eigenmodes of the identical mesh — a credential, not the contribution, and incidentally the first independent validation of the reference stack. Around the capability we kept the honest ledger: no measured performance corner over Palace at scale (the direct eigensolve's advantage is small-to-medium-bounded, with a flop-and-fill crossover below 1M DOFs), a GPU path that is correctness-proven and performance-negative at benchmark sizes, and an eigenmode/EPR branch that is roadmap — blocked at a characterized deep-shift eigensolve wall, with the Jacobi–Davidson-plus-Helmholtz-projection path and the adjoint-eigenpair formulas named. The position we defend is the one the evidence supports: gradient-based electromagnetic design of the transmon's electrostatic Hamiltonian parameters works today, from the field solve itself, on production geometry — and the boundary of that capability is published alongside it.

Acknowledgments. geode-fem's development was AI-orchestrated (agent-driven implementation with human direction and review); this benchmark program and paper follow the same discipline, with every reported number traced to a committed, human-auditable artifact.

References

R. Albanese and G. Rubinacci. Solution of three dimensional eddy current problems by integral and differential methods. *IEEE Transactions on Magnetics*, 24(1):98–101, 1988. doi: 10.1109/20.43865. URL https://doi.org/10.1109/20.43865.

Martin S. Alnæs, Anders Logg, Kristian B. Ølgaard, Marie E. Rognes, and Garth N. Wells. Unified form language. *ACM Transactions on Mathematical Software*, 40(2):1–37, 2014. doi: 10.1145/2566630. URL https://doi.org/10.1145/2566630.

Robert Anderson, Julian Andrej, Andrew Barker, Jamie Bramwell, Jean-Sylvain Camier, Jakub Cerveny, Veselin Dobrev, Yohann Dudouit, Aaron Fisher, Tzanio Kolev, Will Pazner, Mark Stowell, Vladimir Tomov, Ido Akkerman, Johann Dahm, David Medina, and Stefano Zampini. MFEM: a modular finite element methods library. *Computers & Mathematics with Applications*, 81:42–74, 2021. doi: 10.1016/j.camwa.2020.06.009.

Julian Andrej, Nabil Atallah, Jan-Phillip Bäcker, Jean-Sylvain Camier, Dylan Copeland, Veselin Dobrev, Yohann Dudouit, Tobias Duswald, Brendan Keith, Dohyun Kim, Tzanio Kolev, Boyan Lazarov, Ketan Mittal, Will Pazner, Socratis Petrides, Syun'ichi Shiraiwa, Mark Stowell, and Vladimir Tomov. High-performance finite elements with mfem. *The International Journal of High Performance Computing Applications*, 2024. doi: 10.1177/10943420241261981. URL https://doi.org/10.1177/10943420241261981.

Alexander Arsenovic, Julien Hillairet, Jackson Anderson, Henrik Forsten, Vincent Ries, Michael Eller, Noah Sauber, Robert Weikle, William Barnhart, and Franz Forstmayr. scikit-rf: An open source python package for microwave network creation, analysis, and calibration [speaker's corner]. *IEEE Microwave Magazine*, 23(1):98–105, 2022. doi: 10.1109/mmm.2021.3117139. URL https://doi.org/10.1109/mmm.2021.3117139.

AWS Center for Quantum Computing. DeviceLayout.jl: computer-aided design of quantum integrated circuits. https://github.com/aws-cqc/DeviceLayout.jl, 2025. v1.15.0 used for geometry/mesh generation in this work. No journal or arXiv artifact exists as of 2026-07-14 (litsearch absence check).

AWS Quantum Technologies Blog. AWS releases open-source software Palace for cloud-based electromagnetics simulations of quantum computing hardware. https://aws.amazon.com/blogs/quantum-computing/aws-releases-open-source-software-palace-for-cloud-based-electromagnetics-simulations-of-quantum-computing-hardware/, 2023.

Jed Brown, Ahmad Abdelfattah, Valeria Barra, Natalie Beams, Jean-Sylvain Camier, Veselin Dobrev, Yohann Dudouit, Leila Ghaffari, Tzanio Kolev, David Medina, Will Pazner, Thilina Ratnayaka, Jeremy Thompson, and Stan Tomov. libceed: Fast algebra for high-order element-based discretizations. *Journal of Open Source Software*, 6(63):2945, 2021. doi: 10.21105/joss.02945. URL https://doi.org/10.21105/joss.02945.

Tianqi Chen, Thierry Moreau, Ziheng Jiang, Lianmin Zheng, Eddie Yan, Meghan Cowan, Haichen Shen, Leyuan Wang, Yuwei Hu, Luis Ceze, Carlos Guestrin, and Arvind Krishnamurthy. Tvm: An automated end-to-end optimizing compiler for deep learning, 2018. URL https://arxiv.org/abs/1802.04799.

Mingyuan Chi and Shizheng Wen. torch-sla: Differentiable sparse linear algebra with adjoint solvers and sparse tensor parallelism for pytorch, 2026. URL https://arxiv.org/abs/2601.13994.

P. Dular, C. Geuzaine, F. Henrotte, and W. Legros. A general environment for the treatment of discrete problems and its application to the finite element method. *IEEE Transactions on Magnetics*, 34(5):3395–3398, 1998. doi: 10.1109/20.717799. URL https://doi.org/10.1109/20.717799.

Axel M. Eriksson, Lukas J. Splitthoff, Harsh Vardhan Upadhyay, Pietro Campana, Niranjan Pittan Narendiran, Kunal Helambe, Linus Andersson, and Simone Gasparinetti. Automated, physics-guided, multi-parameter design optimization for superconducting quantum devices, 2025. URL https://arxiv.org/abs/2508.18027.

Christophe Geuzaine and Jean-François Remacle. Gmsh: a 3-D finite element mesh generator with built-in pre- and post-processing facilities. *International Journal for Numerical Methods in Engineering*, 79(11):1309–1331, 2009. doi: 10.1002/nme.2579.

Alec M. Hammond, Ardavan Oskooi, Mo Chen, Zin Lin, Steven G. Johnson, and Stephen E. Ralph. High-performance hybrid time/frequency-domain topology optimization for large-scale photonics inverse design. *Optics Express*, 30(3):4467, 2022. doi: 10.1364/oe.442074. URL https://doi.org/10.1364/oe.442074.

Vicente Hernandez, Jose E. Roman, and Vicente Vidal. SLEPc: a scalable and flexible toolkit for the solution of eigenvalue problems. *ACM Transactions on Mathematical Software*, 31(3):351–362, 2005. doi: 10.1145/1089014.1089019.

Ralf Hiptmair and Jinchao Xu. Nodal auxiliary space preconditioning in H(curl) and H(div) spaces. *SIAM Journal on Numerical Analysis*, 45(6):2483–2509, 2007. doi: 10.1137/060660588.

Johannes Hoffmann, Christian Hafner, Patrick Leidenberger, Jan Hesselbarth, and Sven Burger. Comparison of electromagnetic field solvers for the 3d analysis of plasmonic nano antennas, 2009. URL https://arxiv.org/abs/0907.3570.

Philipp Holl, Vladlen Koltun, and Nils Thuerey. Learning to control pdes with differentiable physics, 2020. URL https://arxiv.org/abs/2001.07457.

Miklós Homolya, Lawrence Mitchell, Fabio Luporini, and David A. Ham. Tsfc: A structure-preserving form compiler. *SIAM Journal on Scientific Computing*, 40(3):C401–C428, 2018. doi: 10.1137/17m1130642. URL https://doi.org/10.1137/17m1130642.

Yuanming Hu, Luke Anderson, Tzu-Mao Li, Qi Sun, Nathan Carr, Jonathan Ragan-Kelley, and Frédo Durand. Difftaichi: Differentiable programming for physical simulation, 2019. URL https://arxiv.org/abs/1910.00935.

Tyler W. Hughes, Ian A. D. Williamson, Momchil Minkov, and Shanhui Fan. Forward-mode differentiation of maxwell's equations. *ACS Photonics*, 6(11):3010–3016, 2019. doi: 10.1021/acsphotonics.9b01238. URL https://doi.org/10.1021/acsphotonics.9b01238.

Jiantao Jiang, Yanli Wang, Yifan Wang, and Hehu Xie. Fieldtnn-based machine learning method for maxwell eigenvalue problems, 2024. URL https://arxiv.org/abs/2411.15828.

Sarah Quiñones El Kazdadi. faer: A linear algebra library for the rust programming language. *Journal of Open Source Software*, 11(123):6099, 2026. doi: 10.21105/joss.06099. URL https://doi.org/10.21105/joss.06099.

Yongha Kim, Anthony W. Jung, Sanmun Kim, Kevin Octavian, Doyoung Heo, Chaejin Park, Jeongmin Shin, Sunghyun Nam, Chanhyung Park, Juho Park, Sangjun Han, Jinmyoung Lee, Seolho Kim, Min Seok Jang, and Chan Y. Park. Meent: Differentiable electromagnetic simulator for machine learning, 2024. URL https://arxiv.org/abs/2406.12904.

Robert C. Kirby and Anders Logg. A compiler for variational forms. *ACM Transactions on Mathematical Software*, 32(3):417–444, 2006. doi: 10.1145/1163641.1163644. URL https://doi.org/10.1145/1163641.1163644.

Jens Koch, Terri M. Yu, Jay Gambetta, A. A. Houck, D. I. Schuster, J. Majer, Alexandre Blais, M. H. Devoret, S. M. Girvin, and R. J. Schoelkopf. Charge-insensitive qubit design derived from the Cooper pair box. *Physical Review A*, 76:042319, 2007. doi: 10.1103/PhysRevA.76.042319.

Tzanio V. Kolev and Panayot S. Vassilevski. Parallel auxiliary space AMG for $H(\text{curl})$ problems. *Journal of Computational Mathematics*, 27(5):604–623, 2009. doi: 10.4208/jcm.2009.27.5.013. URL https://doi.org/10.4208/jcm.2009.27.5.013. Crossref records the online-publication year 2018; print year 2009 used here (litsearch cosmetic fix, see version-dir refs.bib header).

Richard B. Lehoucq, Danny C. Sorensen, and Chao Yang. *ARPACK Users' Guide: Solution of Large-Scale Eigenvalue Problems with Implicitly Restarted Arnoldi Methods*. SIAM, 1998. doi: 10.1137/1.9780898719628.

Liang Liang, Minliang Liu, John Elefteriades, and Wei Sun. Pytorch-fea: Autograd-enabled finite element analysis methods with applications for biomechanical analysis of human aorta. *Computer Methods and Programs in Biomedicine*, 238:107616, 2023. doi: 10.1016/j.cmpb.2023.107616. URL https://doi.org/10.1016/j.cmpb.2023.107616.

Qigang Liang, Xuejun Xu, and Qingquan Zhang. Adaptive multilevel methods for the maxwell eigenvalue problem, 2026. URL https://arxiv.org/abs/2603.29718.

Anders Logg and Garth N. Wells. Dolfin: Automated finite element computing. *ACM Transactions on Mathematical Software*, 37(2):1–28, 2010. doi: 10.1145/1731022.1731030. URL https://doi.org/10.1145/1731022.1731030.

Yannik Mahlau, Frederik Schubert, Lukas Berg, and Bodo Rosenhahn. Fdtdx: High-performance open-source fdtd simulation with automatic differentiation. *Journal of Open Source Software*, 11 (117):8912, 2026. doi: 10.21105/joss.08912. URL https://doi.org/10.21105/joss.08912.

J.B. Manges and Z.J. Cendes. A generalized tree-cotree gauge for magnetic field computation. *IEEE Transactions on Magnetics*, 31(3):1342–1347, 1995. doi: 10.1109/20.376275. URL https://doi.org/10.1109/20.376275.

Huan-Ting Meng, Bao-Lin Nie, Steven Wong, Charles Macon, and Jian-Ming Jin. Gpu accelerated finite-element computation for electromagnetic analysis. *IEEE Antennas and Propagation Magazine*, 56(2):39–62, 2014. doi: 10.1109/map.2014.6837065. URL https://doi.org/10.1109/map.2014.6837065.

Zlatko K. Minev, Zaki Leghtas, Shantanu O. Mundhada, Lysander Christakis, Ioan M. Pop, and Michel H. Devoret. Energy-participation quantization of Josephson circuits. *npj Quantum Information*, 7:131, 2021a. doi: 10.1038/s41534-021-00461-8.

Zlatko K. Minev, Thomas G. McConkey, Maika Takita, Antonio D. Corcoles, and Jay M. Gambetta. Circuit quantum electrodynamics (cqed) with modular quasi-lumped models, 2021b. URL https://arxiv.org/abs/2103.10344.

Sean Molesky, Zin Lin, Alexander Y. Piggott, Weiliang Jin, Jelena Vucković, and Alejandro W. Rodriguez. Inverse design in nanophotonics. *Nature Photonics*, 12(11):659–670, 2018. doi: 10.1038/s41566-018-0246-9. URL https://doi.org/10.1038/s41566-018-0246-9.

Jean-Claude Nédélec. Mixed finite elements in $\mathbb{R}^3$. *Numerische Mathematik*, 35(3):315–341, 1980. doi: 10.1007/BF01396415.

Richard B. Nelson. Simplified calculation of eigenvector derivatives. *AIAA Journal*, 14(9):1201–1205, 1976. doi: 10.2514/3.7211. URL https://doi.org/10.2514/3.7211.

Simon E. Nigg, Hanhee Paik, Brian Vlastakis, Gerhard Kirchmair, S. Shankar, Luigi Frunzio, M. H. Devoret, R. J. Schoelkopf, and S. M. Girvin. Black-box superconducting circuit quantization. *Physical Review Letters*, 108:240502, 2012. doi: 10.1103/PhysRevLett.108.240502.

William L. Oberkampf and Christopher J. Roy. *Verification and Validation in Scientific Computing*. Cambridge University Press, 2010. doi: 10.1017/cbo9780511760396. URL https://doi.org/10.1017/cbo9780511760396.

Palace developers, AWS Center for Quantum Computing. Palace: PArallel LArge-scale Computational Electromagnetics. https://github.com/awslabs/palace, 2023. Open-source 3D finite-element solver; v0.13-class, git fba6a5b used in this work. No associated journal article exists as of 2026-07 (absence re-confirmed by litsearch, 2026-07-14).

Greg Peairs. Schematic-driven design of a quantum processor with DeviceLayout.jl. JuliaCon Global 2025, https://pretalx.com/juliacon-2025/talk/7EYFPN/, 2025. Recording: https://www.youtube.com/watch?v=-LohBVtFdwU.

Greg Peairs and Hugh Carson. Design quantum integrated circuits with open-source software DeviceLayout.jl from AWS. https://aws.amazon.com/blogs/quantum-computing/design-quantum-integrated-circuits-with-open-source-software-devicelayout-jl-from-aws/, 2025. AWS Quantum Technologies Blog; source of the transmon+resonator geometry and the reference workflow reproduced here.

Sofia Ponomareva, Adelin Patoux, Clément Majorel, Antoine Azéma, Aurélien Cuche, Christian Girard, Arnaud Arbouet, and Peter R. Wiecha. Torchgdm: A gpu-accelerated python toolkit for multi-scale electromagnetic scattering with automatic differentiation, 2025. URL https://arxiv.org/abs/2505.09545.

Jonathan Ragan-Kelley, Connelly Barnes, Andrew Adams, Sylvain Paris, Frédo Durand, and Saman Amarasinghe. Halide. *Proceedings of the 34th ACM SIGPLAN Conference on Programming Language Design and Implementation*, pages 519–530, 2013. doi: 10.1145/2491956.2462176. URL https://doi.org/10.1145/2491956.2462176.

Taha Rajabzadeh, Zhaoyou Wang, Nathan Lee, Takuma Makihara, Yudan Guo, and Amir H. Safavi-Naeini. Analysis of arbitrary superconducting quantum circuits accompanied by a python package: Sqcircuit. *Quantum*, 7:1118, 2023. doi: 10.22331/q-2023-09-25-1118. URL https://doi.org/10.22331/q-2023-09-25-1118.

Taha Rajabzadeh, Alex Boulton-McKeehan, Sam Bonkowsky, David I. Schuster, and Amir H. Safavi-Naeini. A general framework for gradient-based optimization of superconducting quantum circuits using qubit discovery as a case study, 2024. URL https://arxiv.org/abs/2408.12704.

Florian Rathgeber, David A. Ham, Lawrence Mitchell, Michael Lange, Fabio Luporini, Andrew T. T. Mcrae, Gheorghe-Teodor Bercea, Graham R. Markall, and Paul H. J. Kelly. Firedrake. *ACM Transactions on Mathematical Software*, 43(3):1–27, 2017. doi: 10.1145/2998441. URL https://doi.org/10.1145/2998441.

P. J. Roache. Quantification of uncertainty in computational fluid dynamics. *Annual Review of Fluid Mechanics*, 29(1):123–160, 1997. doi: 10.1146/annurev.fluid.29.1.123. URL https://doi.org/10.1146/annurev.fluid.29.1.123.

Martin F. Schubert, Alfred K. C. Cheung, Ian A. D. Williamson, Aleksandra Spyra, and David H. Alexander. Inverse design of photonic devices with strict foundry fabrication constraints, 2022. URL https://arxiv.org/abs/2201.12965.

Sadman Shanto, Andre Kuo, Clark Miyamoto, Haimeng Zhang, Vivek Maurya, Evangelos Vlachos, Malida Hecht, Chung Wa Shum, and Eli Levenson-Falk. Squadds: A validated design database and simulation workflow for superconducting qubit design. *Quantum*, 8:1465, 2024. doi: 10.22331/q-2024-09-09-1465. URL https://doi.org/10.22331/q-2024-09-09-1465.

David Sommers, Zach Degnan, Divita Gautam, Yi-Hsun Chen, Chun-Ching Chiu, Arkady Fedorov, and Prasanna Pakkiam. Open-source highly parallel electromagnetic simulations for superconducting circuits, 2025. URL https://arxiv.org/abs/2511.01220.

Tracel AI. Burn: a comprehensive deep-learning and tensor-computation framework in Rust. https://github.com/tracel-ai/burn, 2024. v0.21.0 used. No journal/arXiv paper exists (litsearch check 2026-07-14): the repo CITATION.cff lists authors Simard, Fortier-Dubois, Tadjibaev, Lagrange + Burn Framework Contributors, with no DOI and no preferred-citation paper — cited as software.

Shizheng Wen, Mingyuan Chi, Tianwei Yu, Ben Moseley, Mike Yan Michelis, Pu Ren, Hao Sun, and Siddhartha Mishra. Learning, solving and optimizing pdes with tensorgalerkin: an efficient high-performance galerkin assembly algorithm, 2026. URL https://arxiv.org/abs/2602.05052.

Tianju Xue, Shuheng Liao, Zhengtao Gan, Chanwook Park, Xiaoyu Xie, Wing Kam Liu, and Jian Cao. Jax-fem: A differentiable gpu-accelerated 3d finite element solver for automatic inverse design and mechanistic data science. *Computer Physics Communications*, 291:108802, 2023. doi: 10.1016/j.cpc.2023.108802. URL https://doi.org/10.1016/j.cpc.2023.108802.

Jiale Ye, Jiaheng Wang, and Yu-xi Liu. Electromagnetic feature extraction in superconducting quantum circuits: An open-source finite-element workflow using palace, 2025. URL https://arxiv.org/abs/2511.09041.